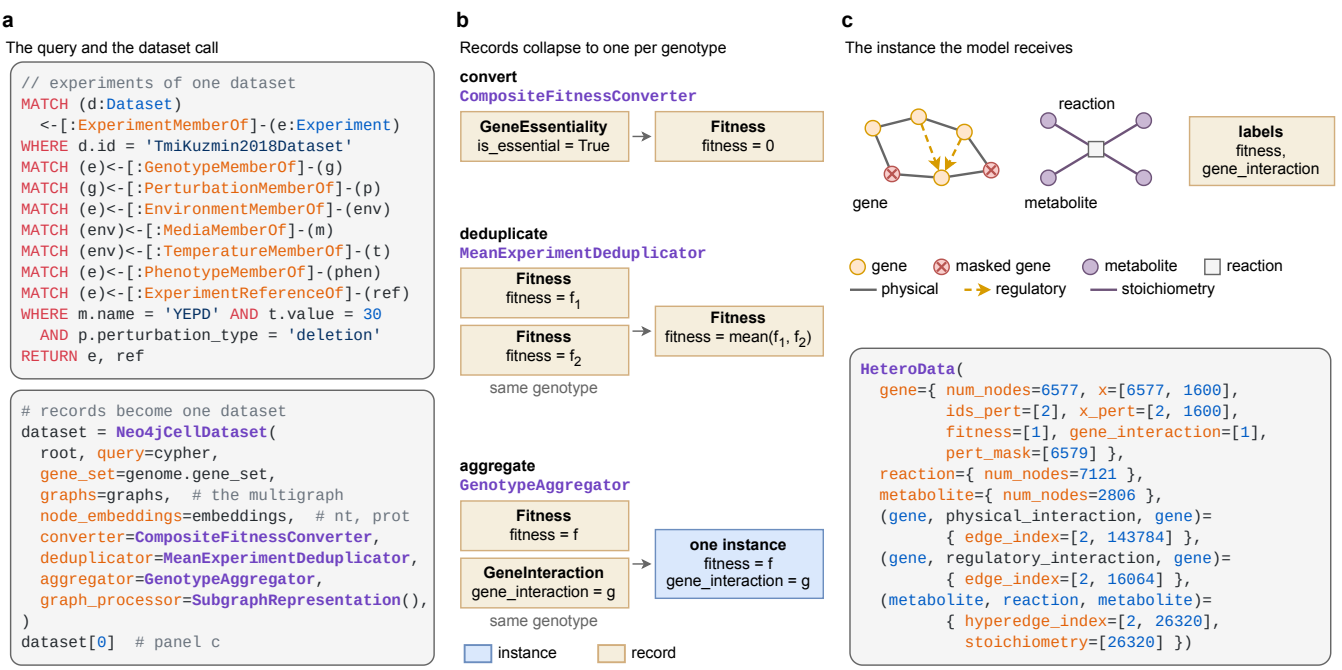

a
The query and the dataset call
// experiments of one dataset
MATCH (d:Dataset)
<-[:ExperimentMemberOf]-(e:Experiment)
WHERE d.id = 'TmiKuzmin2018Dataset'
MATCH (e)<-[:GenotypeMemberOf]-(g)
MATCH (g)<-[:PerturbationMemberOf]-(p)
MATCH (e)<-[:EnvironmentMemberOf]-(env)
MATCH (env)<-[:MediaMemberOf]-(m)
MATCH (env)<-[:TemperatureMemberOf]-(t)
MATCH (e)<-[:PhenotypeMemberOf]-(phen)
MATCH (e)<-[:ExperimentReferenceOf]-(ref)
WHERE m.name = 'YEPD' AND t.value = 30
AND p.perturbation_type = 'deletion'
RETURN e, ref
records become one dataset
dataset = Neo4jCellDataset(
root, query=cypher,
gene_set=genome.gene_set,
graphs=graphs, # the multigraph
node_embeddings=embeddings, # nt, prot
converter=CompositeFitnessConverter,
deduplicator=MeanExperimentDeduplicator,
aggregator=GenotypeAggregator,
graph_processor=SubgraphRepresentation(),
)
dataset[0] # panel c
b
Records collapse to one per genotype
convert
CompositeFitnessConverter
GeneEssentiality
is_essential = True
Fitness
fitness = 0
deduplicate
MeanExperimentDeduplicator
Fitness
fitness = f1
Fitness
fitness = f2
same genotype
Fitness
fitness = mean(f1, f2)
aggregate
GenotypeAggregator
Fitness
fitness = f
GeneInteraction
gene_interaction = g
same genotype
one instance
fitness = f
gene_interaction = g
instance
record
c
The instance the model receives
gene
reaction
metabolite
labels
fitness,
gene_interaction
gene
masked gene
metabolite
reaction
physical
regulatory
stoichiometry
HeteroData(
gene={ num_nodes=6577, x=[6577, 1600],
ids_pert=[2], x_pert=[2, 1600],
fitness=[1], gene_interaction=[1],
pert_mask=[6579] },
reaction={ num_nodes=7121 },
metabolite={ num_nodes=2806 },
(gene, physical_interaction, gene)=
{ edge_index=[2, 143784] },
(gene, regulatory_interaction, gene)=
{ edge_index=[2, 16064] },
(metabolite, reaction, metabolite)=
{ hyperedge_index=[2, 26320],
stoichiometry=[26320] })